RACE Regular Analysis Through Continuous Exercise

Units & Dimensions, Basic Mathematics : RACE-01

1. Which among the following is not a dimension?

(A) Time (B) Kelvin (C) Length (D) Mass

PUV172

2. The SI unit of a physical quantity is pascal-second. The dimensional formula of this quantity will be

(A) $[ML^{-1}T^{-1}]$ (B) $[ML^{-1}T^{-2}]$ (C) $[ML^{2}T^{-1}]$ (D) $[M^{-1}L^{3}T^{0}]$

PUV173

3. The dimensional formula of angular impulse is :

(A) $[M\,L^{-2}\,T^{-1}]$ (B) $[M\,L^{2}\,T^{-2}]$ (C) $[M\,L\;T^{-1}]$ (D) $[M\,L^{2}\,T^{-1}]$

PUV174

4. The de-Broglie wavelength associated with a particle of mass m and energy E is $h/\sqrt{2mE}$. The dimensional formula for Planck's constant is :

(A) $[ML^{-1}T^{-2}]$ (B) $[ML^{2}T^{-1}]$ (C) $[MLT^{-2}]$ (D) $[M^{2}L^{2}T^{-2}]$

PUV175

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5. If G be the gravitational constant and u be the energy density then which of the following quantity have the dimension as that the $\sqrt{uG}$:

(A) Pressure gradient per unit mass (B) Force per unit mass

(C) Gravitational potential (D) Energy per unit mass

PUV176

6. Consider two physical quantities A and B related to each other as $E=\dfrac{B-x^2}{At}$ where E, x and t have dimensions of energy, length and time respectively. The dimension of AB is

(A) $L^{-2}M^{1}T^{0}$ (B) $L^{2}M^{-1}T^{1}$ (C) $L^{-2}M^{-1}T^{1}$ (D) $L^{0}M^{-1}T^{1}$

PUV177

7. The equation of a circle is given by $x^2 + y^2 = a^2$, where a is the radius. If the equation is modified to change the origin other than (0, 0), then find out the correct dimensions of A and B in a new equation: $(x-At)^2+\left(y-\dfrac{t}{B}\right)^2=a^2$.

The dimensions of t is given as $[T^{-1}]$.

(A) $A = [L^{-1}\,T]$, $B = [LT^{-1}]$ (B) $A = [LT]$, $B = [L^{-1}T^{-1}]$

(C) $A = [L^{-1}T^{-1}]$, $B = [LT^{-1}]$ (D) $A = [L^{-1}T^{-1}]$, $B = [LT]$

PUV178

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8. $\left(P+\frac{a}{V^2}\right)(V-b)=RT$ represents the equation of state of some gases. Where P is the pressure, V is the volume, T is the temperature and a, b, R are the constants. The physical quantity, which has dimensional formula as that of $\frac{b^2}{a}$, will be :

(A) Bulk modulus
(B) Modulus of rigidity
(C) Compressibility
(D) Energy density

PUV179

9. Applying the principle of homogeneity of dimensions, determine which one is correct. Where T is time period, G is gravitational constant, M is mass, r is radius of orbit.

(A) $T^2=\frac{4\pi^2 r}{GM^2}$ (B) $T^2=4\pi^2 r^3$ (C) $T^2=\frac{4\pi^2 r^3}{GM}$ (D) $T^2=\frac{4\pi^2 r^2}{GM}$

PUV180

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ANSWER KEY

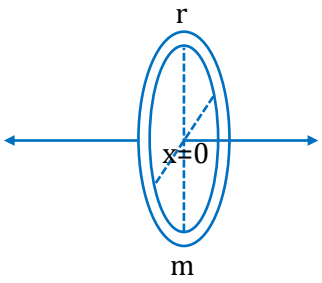
Question	1	2	3	4	5	6	7	8	9
Answer	B	A	D	B	B	B	B	C	C

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RACE
Regular Analysis Through Continuous Exercise

Units & Dimensions, Basic Mathematics : RACE-02

1. A particle is performing SHM along the axis of a fixed ring. Due to gravitational force, its displacement at time t is given by $x = a \sin \omega t$.



In this equation ω is found to depend on radius of the ring (r), mass of the ring (m) and gravitational constant (G).

Using dimensional analysis, find the expression of ω in terms of m, r and G.

PUV181

2. The frequency (ν) of an oscillating liquid drop may depend upon radius (r) of the drop, density (ρ) of liquid and the surface tension (s) of the liquid as $\nu = r^a \rho^b s^c$. The values of a, b and c respectively are:

(A) $\left(-\frac{3}{2}, -\frac{1}{2}, \frac{1}{2}\right)$ (B) $\left(\frac{3}{2}, -\frac{1}{2}, \frac{1}{2}\right)$ (C) $\left(\frac{3}{2}, \frac{1}{2}, -\frac{1}{2}\right)$ (D) $\left(-\frac{3}{2}, \frac{1}{2}, \frac{1}{2}\right)$

PUV182

3. If the velocity of light c, universal gravitational constant G and planck's constant h are chosen as fundamental quantities. The dimensions of mass in the new system is:

(A) $\left[h^{\frac{1}{2}} c^{-\frac{1}{2}} G^{1}\right]$ (B) $\left[h^{1} c^{1} G^{-1}\right]$ (C) $\left[h^{-\frac{1}{2}} c^{\frac{1}{2}} G^{\frac{1}{2}}\right]$ (D) $\left[h^{\frac{1}{2}} c^{\frac{1}{2}} G^{-\frac{1}{2}}\right]$

PUV183

4. If momentum [P], area [A] and time [T] are taken as fundamental quantities, then the dimensional formula for coefficient of viscosity is :

(A) $[P A^{-1} T^{0}]$ (B) $[P A T^{-1}]$ (C) $[P A^{-1} T]$ (D) $[P A^{-1} T^{-1}]$

PUV184

5. The density of a material in SI units is 128 $kg\ m^{-3}$. In certain units in which the unit of length is 25 cm and the unit of mass is 50 g, the numerical value of density of the material is :

(A) 410 (B) 640 (C) 16 (D) 40

PUV185

6. If unit of length and time is doubled, the numerical value of 'g' (acceleration due to gravity) will be:

(A) doubled (B) halved (C) four times (D) remain same

PUV186

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7. If surface tension (S), Moment of inertia (I) and Planck's constant (h), were to be taken as the fundamental units, the dimensional formula for linear momentum would be :-

(A) $S^{3/2}I^{1/2}h^0$ (B) $S^{1/2}I^{1/2}h^0$ (C) $S^{1/2}I^{1/2}h^{-1}$ (D) $S^{1/2}I^{3/2}h^{-1}$

PUV187

8. The characteristic distance at which quantum gravitational effects are significant, the Planck length, can be determined from a suitable combination of the fundamental physical constants G, h and c. Which of the following correctly gives the Planck length?

(A) G^2hc (B) $\left(\frac{Gh}{c^3}\right)^{\frac{1}{2}}$ (C) $G^{\frac{1}{2}}h^2c$ (D) G^2hc^3

PUV188

9. Time (T), velocity (C) and angular momentum (h) are chosen as fundamental quantities instead of mass, length and time. In terms of these, the dimensions of mass would be :

(A) $[M] = [T^{-1}\ C^{-2}\ h]$ (B) $[M] = [T^{-1}\ C^2\ h]$

(C) $[M] = [T^{-1}\ C^{-2}\ h^{-1}]$ (D) $[M] = [T\ C^{-2}\ h]$

PUV189

10. If mass is written as $m = k\ c^P\ G^{-1/2}\ h^{1/2}$ then the value of P will be:
(Constants have their usual meaning with k a dimensionless constant)

(A) 1/2 (B) 1/3 (C) 2 (D) –1/3

PUV190

11. A clock has 75 *cm*, 60 *cm* long second hand and minute hand respectively. In 30 minutes duration the tip of second hand will travel x distance more than the tip of minute hand. The value of x in meter is nearly (Take $\pi = 3.14$)

(A) 139.4 (B) 140.5 (C) 220.0 (D) 118.9

PUV191

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ANSWER KEY

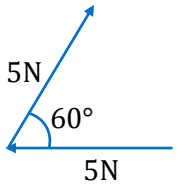
Question	1	2	3	4	5	6	7	8	9	10	11
Answer	ω = (Some number) $\sqrt{\frac{Gm}{r^3}}$	A	D	A	D	A	B	B	A	A	A

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RACE Regular Analysis Through Continuous Exercise

Vectors : RACE-03

1. The forces, each numerically equal to $5N$, are acting as shown in the Figure. Find the angle between forces?



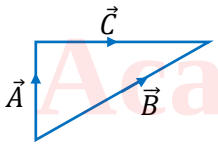
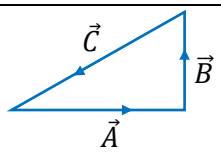
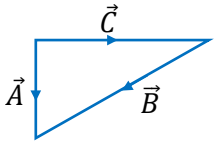
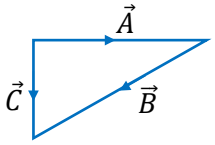
PUV192

2. When vector $\vec{A} = 2\hat{\imath} + 3\hat{\jmath} + 2\hat{k}$ is subtracted from vector $\vec{B}$, it gives a vector equal to $2\hat{\jmath}$. Then the magnitude of vector $\vec{B}$ will be:

(A) $n\sqrt{33}$ (B) 3 (C) $\sqrt{6}$ (D) $\sqrt{5}$

PUV193

3. Match List-I with List-II.

List-I		List-II	
(a)	$\vec{C}-\vec{A}-\vec{B}=0$	(i)	
(b)	$\vec{A}-\vec{C}-\vec{B}=n0$	(ii)	
(c)	$\vec{B}-\vec{A}-\vec{C}=0$	(iii)	
(d)	$\vec{A}-\vec{B}=-\vec{C}$	(iv)	

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Choose the correct answer from the options given below :

(A) (a)→(iv), (b)→(i), (c)→(iii), (d)→(ii) (B) (a)→(iv), (b)→(iii), (c)→(i), (d)→(ii)

(C) (a)→(iii), (b)→(ii), (c)→(iv), (d)→(i) (D) (a)→(i), (b)→(iv), (c)→(ii), (d)→(iii)

PUV194

4. The angle between vector $\vec{Q}$ and the resultant of $(2\vec{Q}+2\vec{P})$ and $(2\vec{Q}-2\vec{P})$ is:

(A) 0° (B) $\tan^{-1}\frac{(2\vec{Q}-2\vec{P})}{2\vec{Q}+2\vec{P}}$ (C) $\tan^{-1}\left(\frac{P}{Q}\right)$ (D) $\tan^{-1}\left(\frac{2Q}{P}\right)$

PUV195

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5. Two forces $\vec{F}_1$ and $\vec{F}_2$ are acting on a body. One force has magnitude thrice that of the other force and the resultant of the two forces is equal to the force of larger magnitude. The angle between $\vec{F}_1$ and $\vec{F}_2$ is $\cos^{-1}\left(\frac{1}{n}\right)$. The value of $|n|$ is _____.

PUV196

6. Two forces having magnitude A and $\frac{A}{2}$ are perpendicular to each other. The magnitude of their resultant is

(A) $\frac{\sqrt{5}A}{4}$ (B) $\frac{5A}{2}$ (C) $\frac{\sqrt{5}A^2}{2}$ (D) $\frac{\sqrt{5}A}{2}$

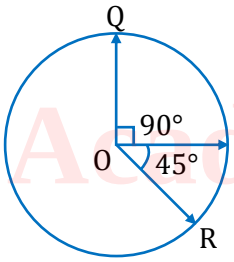
PUV197

7. Two vectors $\vec{A}$ and $\vec{B}$ have equal magnitudes. If magnitude of $\vec{A}+\vec{B}$ is equal to two times the magnitude of $\vec{A}-\vec{B}$, then the angle between $\vec{A}$ and $\vec{B}$ will be :

(A) $\sin^{-1}\left(\frac{3}{5}\right)$ (B) $\sin^{-1}\left(\frac{1}{3}\right)$ (C) $\cos^{-1}\left(\frac{3}{5}\right)$ (D) $\cos^{-1}\left(\frac{1}{3}\right)$

PUV198

8. Three vectors $\overrightarrow{OP}, \overrightarrow{OQ}$ and $\overrightarrow{OR}$ each of magnitude A are acting as shown in figure. The resultant of the three vectors is $A\sqrt{x}$. The value of x is ____.



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PUV199

9. A vector in x-y plane makes an angle of 30° with y-axis The magnitude of y-component of vector is $2\sqrt{3}$. The magnitude of x-component of the vector will be :

(A) $\frac{1}{\sqrt{3}}$ (B) 6 (C) $\sqrt{3}$ (D) 2

PUV200

10. Rain is falling vertically downwards with a speed 5 m/s. If unit vector along upward is defined as $\hat{j}$, represent velocity of rain in vector form.

PUV201

11. A man walks 40 m North, then 30 m East and then 40 m South. Find the displacement from the starting point?

PUV202

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11
Answer	120°	A	B	A	6	D	C	3	D	$\vec{V}_R = -5\hat{j}$	30 m East

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RACE Regular Analysis Through Continuous Exercise

Vectors : RACE-04

1. If the projection of $2\hat{\imath} + 4\hat{\jmath} - 2\hat{k}$ on $\hat{\imath} + 2\hat{\jmath} + \alpha\hat{k}$ is zero. Then, the value of α will be

PUV203

2. If $\vec{a}$ and $\vec{b}$ makes an angle $\cos^{-1}\left(\frac{5}{9}\right)$ with each other, then $|\vec{a} + \vec{b}| = \sqrt{2}|\vec{a} - \vec{b}|$ for $|\vec{a}| = n|\vec{b}|$. The integer value of n is _____.

PUV204

3. Vectors $a\hat{\imath} + b\hat{\jmath} + \hat{k}$ and $2\hat{\imath} - 3\hat{\jmath} + 4\hat{k}$ are perpendicular to each other when $3a + 2b = 7$, the ratio of a to b is $\frac{x}{2}$. The value of x is _____.

PUV205

4. Let $\vec{A} = (\hat{\imath} + \hat{\jmath})$ and $\vec{B} = (2\hat{\imath} - \hat{\jmath})$. The magnitude of a coplanar vector $\vec{C}$ such that $\vec{A}.\vec{C} = \vec{B}.\vec{C} = 1$ is given by

(A) $\sqrt{\frac{5}{9}}$ (B) $\sqrt{\frac{10}{9}}$ (C) $\sqrt{\frac{20}{9}}$ (D) $\sqrt{\frac{9}{12}}$

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PUV206

5. If two vectors $\vec{P} = \hat{\imath} + 2m\hat{\jmath} + m\hat{k}$ and $\vec{Q} = 4\hat{\imath} - 2\hat{\jmath} + m\hat{k}$ are perpendicular to each other. Then, the value of m will be :

(A) 1 (B) –1 (C) – 3 (D) 2

PUV207

6. $\vec{A}$ is a vector quantity such that $|\vec{A}|$ = non-zero constant. Which of the following expressions is true for $\vec{A}$?

(A) $\vec{A}.\vec{A} = 0$ (B) $\vec{A} \times \vec{A} < 0$ (C) $\vec{A} \times \vec{A} = 0$ (D) $\vec{A} \times \vec{A} > 0$

PUV208

7. If $\vec{A}$ and $\vec{B}$ are two vectors satisfying the relation $\vec{A}.\vec{B} = |\vec{A} \times \vec{B}|$. Then the value of $|\vec{A} - \vec{B}|$ will be:

(A) $\sqrt{A^2 + B^2}$ (B) $\sqrt{A^2 + B^2 + \sqrt{2}AB}$

(C) $\sqrt{A^2 + B^2 + 2AB}$ (D) $\sqrt{A^2 + B^2 - \sqrt{2}AB}$

PUV209

8. A force $\vec{F} = (\hat{\imath} + 2\hat{\jmath} + 3\hat{k})$ N acts at a point $(4\hat{\imath} + 3\hat{\jmath} - \hat{k})m$. Then the magnitude of torque about the point $(\hat{\imath} + 2\hat{\jmath} + \hat{k})m$ will be $\sqrt{x}$ N-m. The value of x is ___.

PUV210

For More Join: @IITJEE_Advanced

9. If $\vec{P} = 3\hat{\imath} + \sqrt{3}\hat{\jmath} + 2\hat{k}$ and $\vec{Q} = 4\hat{\imath} + \sqrt{3}\hat{\jmath} + 2.5\hat{k}$ then, The unit vector in the direction of $\vec{P} \times \vec{Q}$ is $\frac{1}{x}\left(\sqrt{3}\hat{\imath} + \hat{\jmath} - 2\sqrt{3}\hat{k}\right)$. The value of x is

PUV211

10. For three vectors $\vec{A} = (-x\hat{\imath} - 6\hat{\jmath} - 2\hat{k})$, $\vec{B} = (-\hat{\imath} + 4\hat{\jmath} + 3\hat{k})$ and $\vec{C} = (-8\hat{\imath} - \hat{\jmath} + 3\hat{k})$, if $\vec{A} \cdot (\vec{B} \times \vec{C}) = 0$, then value of x is _____.

PUV212

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	5	3	1	A	D	C	D	195	4	4

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